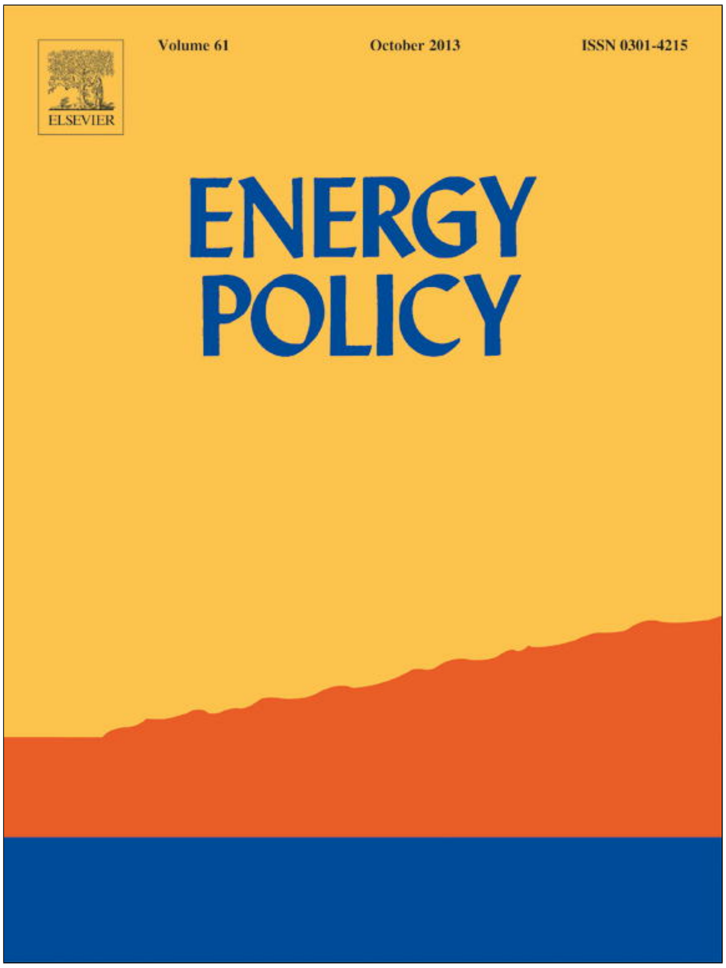
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Household energy consumption patterns and its environmental implications: Assessment of energy access and poverty in Nepal

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HIGHLIGHTS

- Household energy use and air pollutant emissions in Nepal are analyzed based on LEAP framework.
- Household energy use is heterogeneous across the regions and biomass for cooking dominates country's energy-mix.
- Energy Development Index is devised to assess country's energy access and poverty across the regions.
- Scaling up RETs and biomass ICS programs are suggested.
- Coordination with inter-agencies and ODAs is vital in alleviating energy poverty in Nepal.

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ABSTRACT

Approximately 87% of Nepal's total final energy is consumed by households. This paper analyzes the patterns of household energy use and associated air pollutant emissions in Nepal based on LEAP framework for thirteen analytical regions and three end-uses. Four scenarios involving different growth paths for socio economic and energy system development through the year 2040 are considered. The study finds that household energy use is heterogeneous across the regions and biomass for cooking dominates the country's energy-mix. Households' CO_2 emissions are less significant but their local indoor pollutant emissions will continue to rise in the future. To help strengthen government's commitment to the UN's sustainable energy for all initiative, this study devises an energy development index (EDI) to assess country's energy access and poverty across the regions. The results reveal that the current level of both energy access and energy poverty in the country is below the basic human needs and this situation will improve by little in next 30 years. The paper argues that to improve these situations require more coordinated and innovative plans and policies from the government. The paper suggests that greater emphasis will be needed in reducing dependence of biomass for cooking, promoting domestic alternative energy sources, scaling up biomass improved cookstoves programs and developing periodic regional level energy database.

1. Introduction

It is well documented that improving access to affordable and reliable modern forms of energy services is essential especially for developing countries in reducing poverty and promoting economic growth (Leach, 1992; UNDP, 2005; WHO, 2006; UNDP and WHO, 2009; AGECC, 2010; Ekouevi and Tuntivate, 2012). Although the country has made notable progress in recent years, Nepalese people continue to be without basic energy services including access to electricity and clean cooking facilities. For instance, as of 2010, over 30% of the country's total households, around 8 million people, lack access to electricity and 78% of total households rely on traditional biomass for cooking (CBS, 2011). More than 80% of the country's total households are in rural areas (CBS, 2012). In addition, households account for nearly 87% of country's total final energy use (WECS, 2010). This clearly indicates that household energy use, mainly biomass, plays an important role in the energy system and the development of Nepal.

Nepal is one of few developing countries committed to take actions to support the United Nations' *sustainable energy for all* initiative. However, developing any national energy action plans and policies to support this initiative requires a careful understanding of the patterns of household energy consumption in different regions of the country. Several research studies and reports are available that examine household energy issues and its associated environmental and health concerns in Nepal either at aggregate national level or at selected small villages and cities level. While several studies focused on the status and the assessment of renewable energy technologies in Nepal (Pokharel, 2003; Mahat, 2004; Gewali and Bhandari, 2005; Banerjee et al., 2011; KC et al.,

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2011; Gurung et al., 2012; Nepal, 2012a, 2012b; Sapkota et al., 2013), studies by Rijal (1986), de Castro (1995), Mendis and van Nes (1999), Acharya et al. (2005), Bajgain and Shakya (2005), Nepal (2008), Gautam et al. (2009) and Katuwal and Bohara (2009) discussed the prospects and potential of biogas systems in the country. There are also few studies (Joshee, 1986; Amacher et al., 1992; Pokharel et al., 1992; Shrestha, 2002; Heltberg, 2004; Nepal et al., 2011; Link et al., 2012; Sapkota et al., 2012; Singh et al., 2012; AEPC, 2012b, 2013) that highlighted the importance and the challenges of improve cookstoves (ICS) and cooking fuel choice in the rural areas of Nepal. Furthermore, there are many studies related to exposure from indoor air pollutant emissions associated with combustion of fuels for cooking and heating in Nepal (Pandey, 1984; Pandey et al., 1985, 1989, 1990; Melsom et al., 2001; Pokhrel et al., 2005, 2013; Shrestha and Shrestha, 2005; WIN, 2005; Joshi et al., 2009; Dhimal et al., 2010; Malla et al., 2011; Adhikari, 2012; Pant, 2012; Singh et al., 2012; Bates et al., 2013; Gurung and Bell, 2013; Kurmi et al., 2013). Most of these studies find that indoor pollution concentrations considerably higher than the common health-based guideline values in the country and exposure to these emissions causing significant negative health impacts, particularly among women and children. Moreover, based on fuel combustion and emission factors, studies by Shrestha and Malla (1996), Shah and Nagpal (1997), Shrestha and Rajbhandari (2010) and Pradhan et al. (2012) estimated sector-wise emissions of local air pollutants including household sector for Kathmandu valley.

More specifically, there are many research studies on Nepal's household energy use in the literature. For instance, Pokharel et al. (1992) and Pokharel (2004) studied energy use for cooking at the national level. Several studies (Rijal et al., 1990; Amacher et al., 1993, 1996; Amatya et al., 1993; Sharma and Bhattacharya, 1997; Link et al., 2012; Baland et al., 2013; Gurung and Oh, 2013; Pant, 2013) analyzed traditional biomass use in the households particularly in the rural areas of Nepal. Also, several overseas development agencies (ODA) have published research reports on household energy-related issues in the country (World Bank, 1985; WIN, 2005; UNDP, 2007, 2012; ADB, 2010a,b; Banerjee et al., 2011). Recently, studies by Practical Action (2009) and Parajuli (2011) illustrated the situation of energy poverty in the country. While the study by Practical Action (2009) discussed a short-term plan of how energy poverty could be reduced in next 10–15 years at the aggregate level, Parajuli (2011) discussed the energy access situation in Mid/Far West development region of the country. In addition, UNDP and WHO (2009), GoN and UNDP (2011) and Bazilian et al. (2012) highlighted issues concerning access to modern energy services in Nepal. However, comprehensive analysis of the patterns of household energy use and its environmental implications at the regional level in the country has received relatively less attention. Analyzing these issues is important and necessary to make in-depth assessment of energy access and poverty situation in the country. This paper analyzes the patterns of household energy consumption at 13 analytical levels and for 3 different end-uses (i.e., cooking, lighting and use of appliances) in Nepal for the base year 2010 and for the future years. Altogether, four scenarios, involving different growth paths for socio economic and energy system development through the year 2040, are considered. This study also makes detailed assessment of access to modern energy services[1] and energy poverty[2] situation in the country. Furthermore, to understand how different regions of the country compare in terms of their energy access and poverty, this study devises Energy Development Index (EDI)[3] for Nepal. The author is not aware of any studies in the literature that developed region-wide EDI in assessing energy poverty in Nepal. The findings presented in this paper may help the government of Nepal in designing effective and innovative national energy actions plans to alleviate energy poverty in the country. This paper may also serve as a reference source for organizations that are involved in household energy sector development programs in Nepal and in other developing countries of the world.

The paper is divided into six sections. Section 2 discusses a brief overview of Nepal's current socio-economic situation and the patterns of household energy use. Section 3 gives a brief description of data and data sources used in the study. Section 4 presents the methodological framework followed by scenarios description in Section 5. Analyses of results and key findings are presented in Section 6. The final section provides the conclusion.

2. Overview of Nepal's socio-economic and household energy use

As of 2011, Nepal's population is 26.5 million and the number of households is about 5427 thousand (CBS, 2012). Geographically, Nepal is divided into three distinct east-west ecological regions: mountains (the Himalayas) in the north, *tarai* (the plains) in the south, and hills in between. For administrative purposes, the country is divided into five development regions: Eastern, Central, Western, Mid-western and Far-western. The *tarai* region is home for half of the population while hills is home for 43% of the population and mountains is home for remaining 7% of the population (Table 1). Only about 17% of total population and 20% of total households are in urban areas. Within five development regions, Central development region holds the highest population (36.5%). This region is also home to 30% of total rural population and 64% of total urban population. On average, urban household income is almost twice the income of rural household. Between two census periods, country's total population is increased by 15%, while the number of households is increased by much higher percentage at 33%. Overall, Nepal's total population grew by 1.4% per year from 2001 to 2011. Economy-wise, Nepal has experienced a marked slowdown in growth during the 2000s, averaging less than 4% annually. The share of both agriculture and industry sectors in country's total gross domestic product (GDP) has decreased over the last two decades (NPC, 2011). However, the service sector has grown and this sector's activities account for a large share of Nepal's economy. For example, in 2010, services accounted for 50% of GDP, agriculture for 36% and industry for 14%. Industrial activity in the country mainly involves the processing of agricultural products.

In terms of energy resources in Nepal, they are broadly classified into three categories – traditional, commercial and alternative. Traditional sources include biomass (i.e., fuelwood, agriculture resides and animal dung), commercial sources include fossil fuels and electricity, and alternative sources include new and renewables. Nepal has no significant reserves of fossil-fuel resources. All petroleum products and over 75% of coal are imported from India (WECS, 2010). Natural gas is not used in the

[1] In the literature, there are several definitions of “modern energy access”. In this study, access to modern energy services is defined as the combination of household's access to electricity and clean cooking facilities. This excludes electricity access to commercial and public buildings that are crucial to country's economic and social development.

[2] Energy poverty is inversely related to access to modern energy services, although improving access is only one of several factors in efforts to reduce energy poverty. There are different approaches to measuring energy poverty (Pachauri et al., 2004; Khandker et al., 2012; Nussbaumer et al., 2012; Sovacool et al., 2012). In this study, we use energy development index (EDI) to measure Nepal's energy poverty.

[3] The EDI, which is developed by International Energy Agency (IEA), is an indicator often used in measuring progress in the transition to cleaner cooking fuels and the degree of maturity of energy end use.

Table 1
Estimated population, number of households and household characteristics by analytical region * (2011).

Analytical region	Population (%)	No. of households (%)	Average size of dwelling (Sq. ft.)	Average household income (US$)	% change (2001–2011)	
					Population	No. of households
Urban**	17.1	19.8	571	4402	40.2	33.1
Mountains	0.3	0.4	651	2352	100.4	126.7
Hills—Kathmandu valley	5.2	6.0	555	5597	38.0	55.3
Hills—Others	4.0	4.6	558	4159	58.3	78.2
Tarai	7.6	8.8	589	3683	32.0	70.5
Rural	82.9	80.2	614	2379	10.9	26.7
Mountains	6.4	6.3	651	2204	3.9	15.5
Hills—Eastern	7.7	8.5	628	2285	0.1	20.5
Hills—Central	11.3	12.0	705	2626	9.7	32.7
Hills—Western	6.5	7.4	619	2186	−0.7	23.4
Hills—Mid/Far western	8.4	8.2	535	1696	9.7	26.6
Tarai—Eastern	9.6	8.9	582	2403	11.2	23.1
Tarai—Central	14.3	12.6	641	2631	21.9	35.6
Tarai—Western	8.2	7.7	612	3099	10.3	26.0
Tarai—Mid/Far western	10.5	8.6	526	2204	21.9	29.3
Nepal	100.0	100.0	605	2800	15.0	33.1

* Nepal is divided into 14 zones and 75 districts. These administrative districts are further divided into smaller units, called village development committees (VDCs) and municipalities. The VDCs are rural areas and municipalities are urban areas. Currently, there are 3915 VDCs and 58 municipalities. These 58 municipalities include 1 metropolitan city (Kathmandu, the capital city of Nepal), 4 sub-metropolis and 53 municipalities. All together, 13 regions are considered in the study. The third nation-wide household survey (CBS, 2012) considers 12 analytical regions. This study considers 13 analytical regions by extending mountains region to urban- and rural- mountains.

** Urban population includes population of 58 municipalities and additional 41 new municipalities proposed by the government. Except for Kathmandu municipality, urban population of each analytical regions in year 2011 (P_i^t) is estimated by using the following expression: $P_i^t = \sum_j P_{i,j}^0 (1+r)^n + \sum_k P_{i,k}^t$; $k,j \in i$ and $k \neq j$, where $P_{i,j}^0$is the urban population of analytical region i and existing municipality j in year 2001,$P_{i,k}^t$is the urban population of analytical region i and new municipality k in year 2011, and r is the average annual growth rate (AAGR) between 2001 and 2011. The urban population of Kathmandu municipality in year 2011 is based on country's urban AAGR of 3.44%. Likewise, urban population for 41 new municipalities in year 2011 is assumed as 10000 for each municipality that are in mountains region and 20000 for each municipality that are in hills and *tarai* regions. For simplicity, national average urban household size (i.e., 4.05) is assumed for all urban analytical regions. The corresponding rural population in year 2011 is then estimated by subtracting urban population from national urban total for each analytical region.
Sources: CBS (2010, 2011, 2012), HNS (2012).

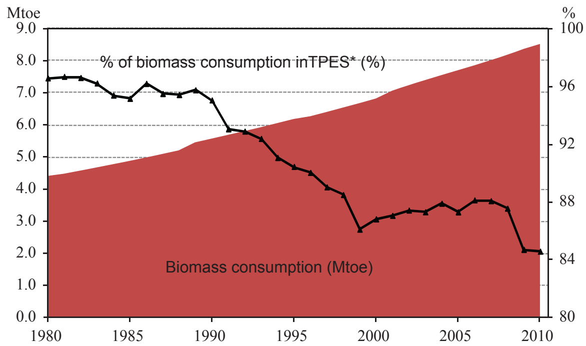

Fig. 1. Primary solid biomass consumption in Nepal. *Note*: *TPES is total primary energy supply. .
Sources: WECS (2010), NEA (2011), IEA (2012a), MoF (2012), NOC (2013)

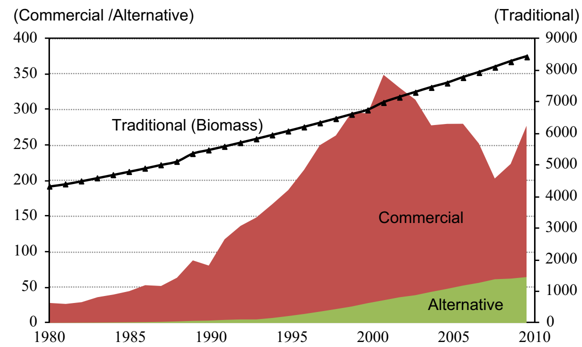

Fig. 2. Energy consumption in the households by energy types in Nepal (ktoe). .
Sources: WECS (2010), NEA (2011), IEA (2012a), MoF (2012), NOC (2013)

country. However, Nepal has enormous potential for hydropower and the country is generously endowed with new and renewable energy resources, including biogas, solar, wind and geothermal. Despite country's enormous hydropower potential, so far less than 1% has been developed (NEA, 2011). Within different types of energy sources, biomass dominates country's energy-mix. The proportion of biomass in Nepal's total primary energy supply (TPES), around 85% in 2010, is considerably higher than most of the developing countries of the world. Although its proportion in TPES has decreased over time, demand for biomass continues to rise in absolute values (Fig. 1). For example, consumption of biomass, most of it by rural households, increased from 4.4 Mtoe in 1980 to 8.5 Mtoe in 2010. Within biomass, fuelwood remains the dominant fuel as it accounted for about 89% of total biomass supply in the 1990s and early 2000s.

Sector-wise, the household sector dominated the total final energy demand. When biomass is included, energy demand by households accounted for almost 87% of total energy demand in 2010 but this sector accounted for only 21% of total energy if biomass is excluded. This reflects wide differences in the consumption pattern of rural and urban households as most of the rural households in Nepal depend on biomass for their cooking needs (Table A1). Due to progressive rise in income and urbanization, biomass fuels are replaced by more efficient fuels – liquefied petroleum gas (LPG), biogas and electricity – primarily for cooking and lighting. As a result, household energy consumption, excluding biomass, grew much faster at 8.7% per year, while it grew overall at 2.4% per year during 1980–2010 (Fig. 2). However, biomass, in particular fuelwood, is expected to remain the primary fuels for the households in a foreseeable future. In per capita terms, Nepal's TPES is increased by only 11% between 2000 and 2010, mainly due to population expanding by 1.4% annually during the period. Although total electricity consumption per capita is almost doubled from 62 kWh in 2000 to 104 kWh in 2010, it remains extremely low when compared to other countries of the world. For example, country's per capita electricity consumption in

2010 is less than 4% of world average in 2010 and it is where India and Pakistan were in 1974, Sri Lanka in 1981 and Bangladesh in 2000 (World Bank, 2012). Many factors influence country's low energy use and electricity consumption per capita behind that of other Asian developing countries, including slow transition to political stability after a decade long civil war and poor electricity supply infrastructure in the country.

Household energy use patterns are closely related to the levels of energy access and poverty. Recently, these issues have received global attention particularly in developing countries. This is evident from the initiative by the United Nations in declaring 2012 as the year of sustainable energy for all. In this regards, Nepal has made notable progress in recent years in improving access to energy services. For example, share of country's total households that have access to electricity almost doubled from 37% in 2004 to 70% in 2010 (CBS, 2004, 2011). Furthermore, by the end of 2011, more than 258 thousand biogas plants and about 620 thousand ICS are disseminated through various government programs mainly to rural households across the country (AEPC, 2012b). Despite these encouraging statistics, Nepal's energy poverty level remains one of the highest in the world. For comparison, country's EDI of 0.091 is at the bottom of rankings compared to selected developing countries of Asia and other regions of the world, reflecting country's low rates of household electrification and high share of biomass use for cooking (Fig. 3).

There are also significant variations in energy poverty level region-wise and between urban and rural areas in the country. For instance, Kathmandu valley has the highest level of EDI among all the regions in the country including all four indicators of the regions (Table 2). The valley is followed by urban hills and urban *tarai* regions in descending order. However, the rest of the regions including urban mountains have dismissal values of EDI. Rural *tarai* of Eastern development region and rural hills of Central development region have the lowest energy development indices among all the regions. This is because most of the households in these regions lack access to electricity and they rely heavily on biomass for cooking and heating.

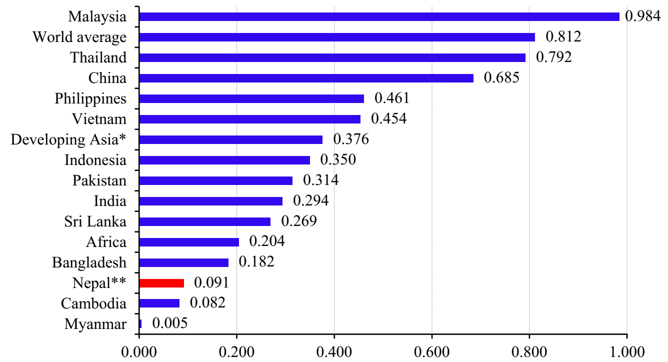

Fig. 3. Energy Development Index (EDI)*** of selected developing countries and regions of the world in 2009. *Notes*:*Developing Asia excludes China. **Population data is based on CBS (2012). ***EDI is calculated as the arithmetic mean of each index created from four indicators, namely, per capita commercial energy consumption, per capita electricity consumption in the household sector, share of modern fuels in total household sector energy use and share of population with access to electricity. Each indicator is calculated using the formula: Indicator=(actual value−minimum)/(maximum value−minimum value). .
Sources: IEA (2011, 2012a), CBS (2012), World Bank (2012)

3. Data and data sources

Data used in this study are collected from various official and international organizations publications. At the national aggregate level of household energy use, data on traditional biomass are employed from periodic energy synopsis reports by Water and Energy Commission Secretariat (WECS, 2006, 2010) and economic survey by Ministry of Finance (MoF, 2012). Data on commercial household fuels are employed from imports and sales of petroleum products data by Nepal Oil Corporation (NOC, 2013), annual reports by Nepal Electricity Authority (NEA, 2009, 2010, 2011) and energy statistics by IEA (IEA, 2012a). In addition, data on alternative household fuels are employed from renewable energy data book by Alternative Energy Promotion Center (AEPC, 2012a). While these publications report data on different energy sources in different measurement units, all values are converted to the common energy units of kilo ton of oil equivalent (ktoe) using caloric and other conversion factors reported in WECS and IEA. Likewise, data on both national and region-wide population and number of households are based on national population and housing census 2011and the past census data published by Central Bureau of Statistics (CBS, 2010, 2012). For the purpose of this study, none of these publications provide complete dataset. Therefore, some of these missing data in the base year 2010 are extrapolated, in particular, household energy are based on its historical data from 2007 to 2009, after the post political conflict in the country.

At the disaggregate level, data from CBS's third Nepal Living Standards Survey (NLSS III) 2010/2011 statistical report (CBS, 2011), Ministry of Population and Health's (MoPH) 2011 Nepal Demographic and Health Survey (MoHP et al., 2012), the World Bank's joint renewable energy study with AEPC (Banerjee et al., 2011), WECS's energy synopsis report (WECS, 2010), ADB's Nepal's power sector report (ADB, 2004) and the local news article by the HNS (2012) are used to determine region-wide detailed energy requirements of households and their socio-economic and demographic characteristics. Periodic NLSS household surveys are widely used as a source of data for various purposes. These data

Table 2
Energy development index (EDI) by analytical regions in Nepal (2010).

Analytical region	Electricity use for appliances index	Total electricity use index	Clean cooking fuel index	Access to electricity index	EDI
Mountains	0.019	0.049	0.037	0.434	0.135
Urban-hills—Kathmandu valley	1	1	1	1	1
Urban-hills—others	0.793	0.707	0.142	0.961	0.651
Urban—*tarai*	0.779	0.675	0.121	0.898	0.618
Rural-hills—Eastern	0.000	0.042	0	0.434	0.119
Rural-hills—Central	0.003	0.090	0.005	0.283	0.095
Rural-hills—Western	0.005	0.126	0.013	0.541	0.171
Rural-hills—Mid/Far western	0.004	0	< 0.001	0.655	0.165
Rural-*tarai*—Eastern	0	0.042	0.008	0	0.013
Rural-*tarai*—Central	0.003	0.090	0.002	0.702	0.199
Rural-*tarai*—Western	0.005	0.126	0.020	0.670	0.205
Rural-*tarai*—Mid/Far western	0.004	0.119	0.014	0.706	0.211
Nepal	0.168	0.214	0.037	0.613	0.258

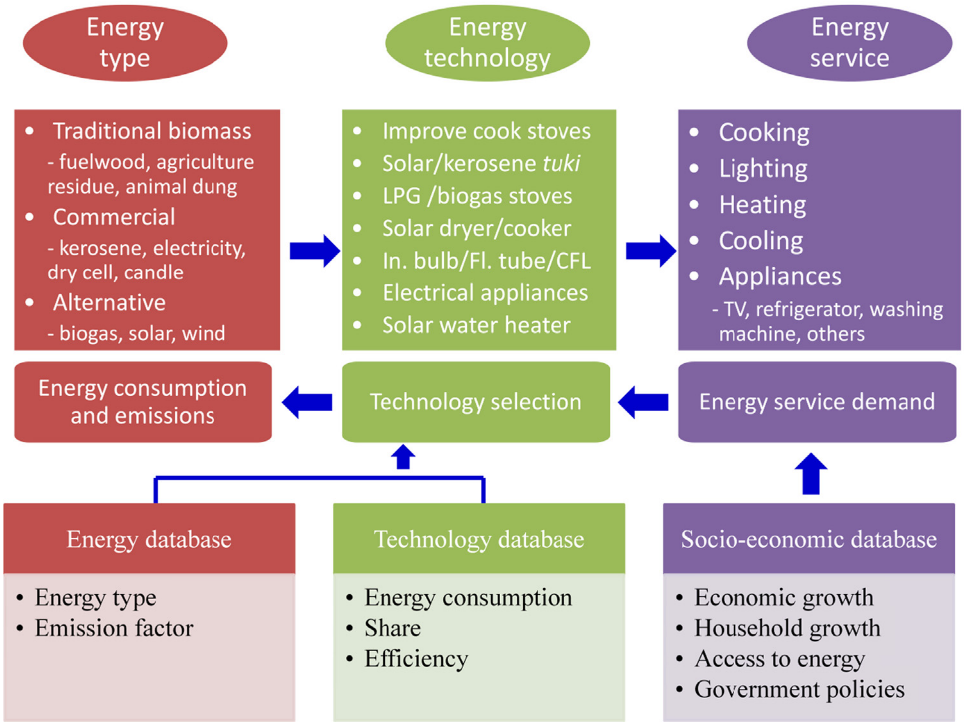

Fig. 4. Structure of Nepal's household sector energy system modeling used in LEAP.

consist of multiple topics related to household welfare including region-wide demography that includes population, household size and dwelling area. However, energy appears only as household energy- and cookstoves- use for cooking, and access to electricity in the survey. Likewise, research studies by the World Bank, United Nations Development Programme (UNDP) and Asian Development Bank (ADB) provide household energy use patterns by end-uses. Relevant information is extracted from these surveys for this study.

4. Methodological framework

In the literature, there are wide varieties of methods used in energy demand analysis. A review study by Bhattacharyya and Timilsina (2010) finds that two types of approaches, econometric and end-use accounting, are commonly used in the existing demand models in developing countries. This study uses LEAP, the Long range Energy Alternatives Planning System, to analyze Nepal's household sector energy demand and corresponding emissions, and to access energy services in the future. LEAP is a comprehensive integrated scenario-based energy-environment model building tool to estimate the future energy demand and emissions at the local-, national-, and regional- levels (Heaps, 2012). The analysis uses a straightforward end-use accounting methodology in which household energy demand of different fuels are calculated as the product of an activity level measuring the level of energy service provided and an energy intensity. Levels of activity in the household sector is first projected forward based on overall assumptions about levels of growth of the Nepalese economy and how its household structure might shift (e.g., household size, rural–urban transition and energy efficiency) as income levels grow. In the model, these flows of energy and materials are character characterized through detailed representation of technologies providing an end-use scenario-driven analysis. The scenario is driven forward by two high level exogenous assumptions: population and average income levels. For population, the UN population projection is used (UN, 2012) and average income levels are assumed to continue to grow in Nepal. The simplified structure of the LEAP modeling framework of Nepal's household sector is presented in Fig. 4. For the purpose of energy projections, the end-use demands in the household sector are disaggregated into lighting, cooking and appliances.

The database used in the model includes details of energy types, energy technologies, energy-related services and counter measures. The energy database comprise of different energy types and their prices and maximum availability as its parameters while the energy technology database comprise of details corresponding to life time energy consumption per unit of output and the availability factor. Likewise, the energy service database considers the demand for each service in the household sector. Based on the energy consumption, emissions are estimated. The database comprise of input parameters that represent each stage of an energy system including conversion of primary energy to secondary energy and secondary energy consumption.

In this study, the energy system of the household sector is broadly classified into two components: energy supply and service demand. Energy supply represents both primary and secondary energy sources. For cooking and heating activities, altogether 10 existing and new technologies options are considered. Likewise, for lighting, 9 existing and new technologies options and altogether 8 electric appliances are considered. A survey by CBS (2011) is used as the basis for classification of the household sub-sectors of the service demand (Fig. 5). The activity parameters, exogenous inputs to the model, are used to estimate the future service demand.

5. Scenario description

The state of country's household sector energy demand situation for the next 30 years can be broadly visualized as a combination of market integration (i.e., extent of liberalization and economic growth) and the nature of governance (i.e., centralization vs. decentralization).

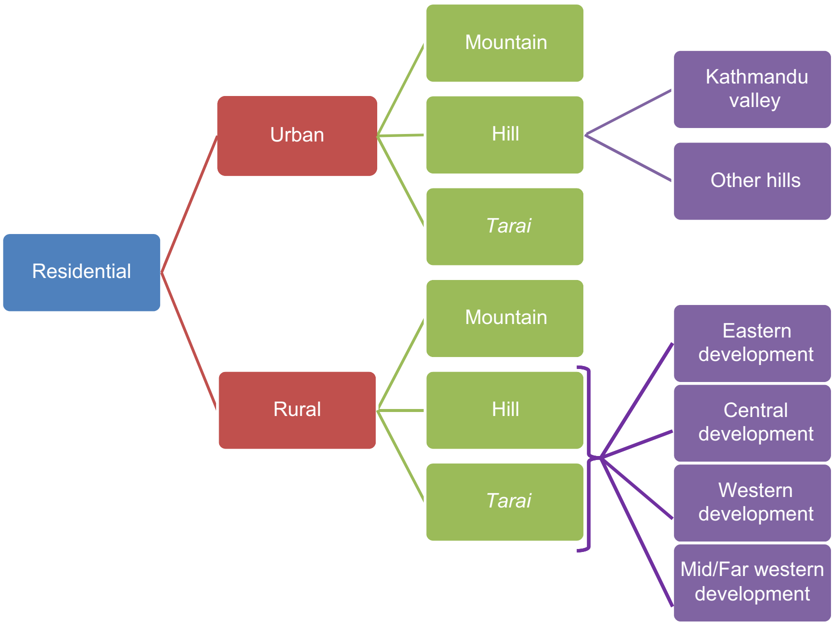

Fig. 5. Classification of the household sector in Nepal used in LEAP.

These two dimensions are similar to those considered in the Intergovernmental Panel on Climate Change's Special Report on Emissions Scenarios (SRES) report and they are adapted to the Nepali context IPCC (2000). Altogether four scenarios are presented in the study: the high economic growth (HIG) scenario, the business-as-usual (BAU) scenario, the sustainable energy development (SED) scenario and the low carbon society (LCS) scenario. The characterization of these four scenarios is illustrated in Fig. 6. The starting year is 2010 and the projection period runs to 2040. Population growth is one of the key drivers of future the household sector energy trends. Depending on the scenarios, country's population grows from 26.5 million in 2011 to around 34.4–37.6 million in 2040 (UN, 2012). In line with the long-term historical trend, the population growth rate slows progressively over the projection period. Socio-economic and technological changes are the other main driving factors in projecting the household sector energy demand. Although the magnitude of changes vary in each scenario, these changes include increase in household incomes, migration from rural housing to more efficient urban housing, progressively declining population growth rate, more use of energy efficient equipments and government plan and policies. In addition, assumptions of future technological development as well as the future economic development are also made for each scenario. No new technologies for cooking, lighting and electrical appliances are assumed. However, existing cooking and lighting end-use technologies would become steadily more energy efficient over the study period. The pace of end-use efficiency gains measured in changes in energy intensity varies for each scenario. Due to the huge amount of data and information included in the quantitative formulation of the scenarios, a brief quantitative summary of the four scenarios are presented in Table A2.

5.1. Business-as-usual (BAU) scenario

The BAU scenario, defined as the reference scenario, is our central scenario. This scenario is based on historical trends and it takes account of current government plans and policies. In this scenario, country follows closely the national development plans and policies. The economic growth is moderate and the population projection would be average based on the UN medium variant projection, but urban to rural migration in the country would rise. In addition, moderate energy efficiency improvement of both cooking and

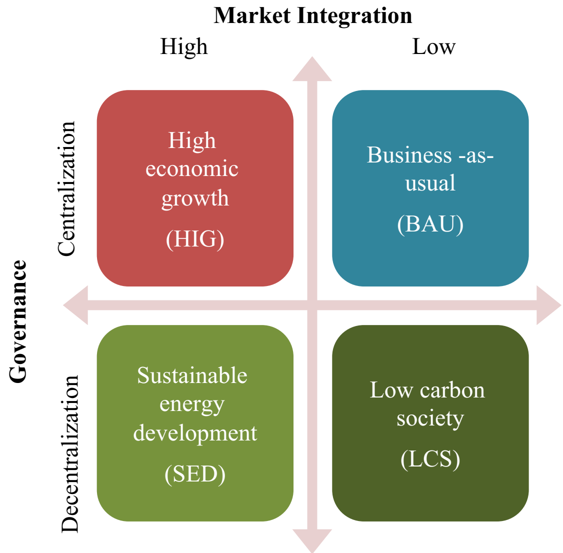

Fig. 6. Scenario description used in the study.

lighting devices is proposed. These efficiency improvements are reflected in decrease in energy intensity of cooking. Despite the use of more efficient lighting devices, the energy intensity of lighting would increase due to increase in both household income and access to electricity. The increase in share of modern fuels for cooking would be moderate.

5.2. High economic growth (HIG) scenario

The HIG scenario is characterized by Nepal being more and more integrated into global markets that lead to high economic growth and there is a faster transition of the economy from agriculture based economy towards industry- and commerce-based economy. Because of easy access to global technology market and modernization of economic sectors under this scenario, significant improvement in the efficiency of cooking and lighting end-use devices is considered. In the demographic front, the country's average annual population growth is low due to low total fertility rate based on the United Nations low variant population projection over the study period. Compared to 2010, the urbanization rate would double in 2040 due to migration from rural to urban areas and transformation of big villages into cities. Demand for modern energy sources would be high in order to sustain high economic growth. The priority is more on the economic growth than on the conservation of environment. Overall, in this scenario, the general development mode of Nepali economy is completely market-driven.

5.3. Sustainable energy development (SED) scenario

The SED scenario focuses and supports activities that promote sustainable development. The country's economic growth is considerable. However, the population growth would be the same as that of BAU scenario. Clean energy and clean production would be the mainstream of the society in this scenario. Rural energy and electricity development would be the government priorities. Accordingly, a higher share of renewable energy technologies is introduced. The energy efficiency improvement for cooking and lighting devices would be high. This scenario is a consequence of active polices towards balanced economic and social development in the country.

5.4. Low carbon society (LCS) scenario

The LCS scenario is based on the necessity of strong local communities. The economic growth in the country is at low level and the population growth is moderate based on the United Nations medium variant projection. There is also low energy efficiency improvement. The households would have low home appliances ownership. However, shifting from fossil-fuels to domestic energy resources primarily biomass would be given the priorities. Local capacity building and importance of self-reliance would prevail.

6. Results and discussion

6.1. Household energy use patterns

The household sector accounted for most of the energy consumed in the country in the past and this trend is projected to remain the same in next 30 years. If the current trend continues, household energy demand is projected to increase from 8765 ktoe in 2010 to 11852 ktoe in BAU scenario in 2040, an increase of 22% or an average growth of 1.1% per year (Table 3). While demand increases consistently throughout the study period, the growth rate slows from an average of 1.2% per year in 2010–2025 to 0.9% per year in 2025–2040. This shift is attributed largely to a progressive decline in population growth, increase in the share of modern fuels for cooking and improved levels of energy efficiency. With the exception of LCS scenario, household energy demand increases more quickly in the BAU scenario than in HIG and SED scenarios. These two scenarios (HIG and SED) have very modest household energy demand growth between 2010 and 2040, increase of 5% in HIG scenario and 9% in SED scenario. In 2040, energy demand in the HIG scenario is 17% lower than in the BAU scenario due mainly to the lower population growth and higher share of modern fuels for cooking. In contrast, energy demand under the LCS scenario increases the most at 11390 ktoe in 2040, an increase of 30% in 30 years due to higher level of biomass consumption.

Despite the rapidly growing availability of modern fuels in recent years, in particular, electricity and LPG, biomass remains the primary energy sources of households in Nepal. Although the share of biomass in total household energy demand in the country has been declining over the last 15–20 years, their share is expected to remain high in next 30 years. More than four-fifth of household sector energy demand will come from biomass, mainly fuelwood, in all four scenarios. For example, the share of biomass in total household energy demand varies from 82% in SED scenario to 87% in BAU scenario in 2040. On the other hand, the outlook of modern fuels differs across the scenarios. Between 2010 and 2040, demand for modern fuels increases from average of 3.9% per year in the LCS scenario to average of 4.9% per year in the HIG scenario. Within the modern fuels, the growth in demand for electricity and LPG are the strongest of all energy sources. The demand for both electricity for lighting and LPG for cooking reaches roughly five-folds in 2040. However, demand for kerosene, used primarily for lighting in the rural areas, declines in all scenarios due mainly to removal of government subsidies for end-users. Although the share of biogas in total energy mix remain low in next 30 years, its average annual growth in demand is the most of all fuel types ranging from 6.7% in HIG scenario to 9.2% in LCS. The outlook of solar in the household energy mix in absolute terms in all scenarios remains less significant over the study period.

Cooking is by far the most important energy user in the household sector of Nepal. Almost 94% of urban- and 99% of rural-household energy consumption is used for cooking activities in 2010 (Fig. 7). In contrast, the share of remaining two end-uses (i.e., lighting and electric appliances use) in urban and rural household energy use is less significant. These two activities together accounts for only 54 ktoe, about 6%, of total urban household energy use and 73 ktoe, about 1%, of total rural household energy use in 2010. This low share is attributed to limited access to reliable and affordable electricity and modern fuels in rural areas, unreliable supply of electricity in urban areas and more importantly, lack of energy infrastructure and development in the country. Despite the low shares of these two end-uses, demand for energy in next 30 years is projected to grow most rapidly at

Table 3
Estimated household sector energy demand under different scenarios for selected years (ktoe).

Energy source	2010	BAU		HIG		SED		LCS	
		2025	2040	2025	2040	2025	2040	2025	2040
Biomass	8421	9632	10267	8288	7616	8468	7826	9278	9614
Modern fuels	273	541	1073	613	1144	540	1067	454	848
Of which: LPG	98	238	491	277	532	235	453	220	437
Kerosene	35	36	29	34	25	9	–	20	–
Electricity	102	248	553	285	587	291	615	195	411
Charcoal	38	19	–	17	–	6	–	19	–
Alternative fuels	65	240	507	234	461	282	665	388	920
Of which: Biogas	65	240	507	234	461	281	661	387	915
Solar	< 1	< 1	< 1	< 1	< 1	1	4	1	5
Miscellaneous fuels	5	4	4	4	3	2	–	6	7
Total	8765	10417	11852	9140	9226	9292	9558	10126	11390

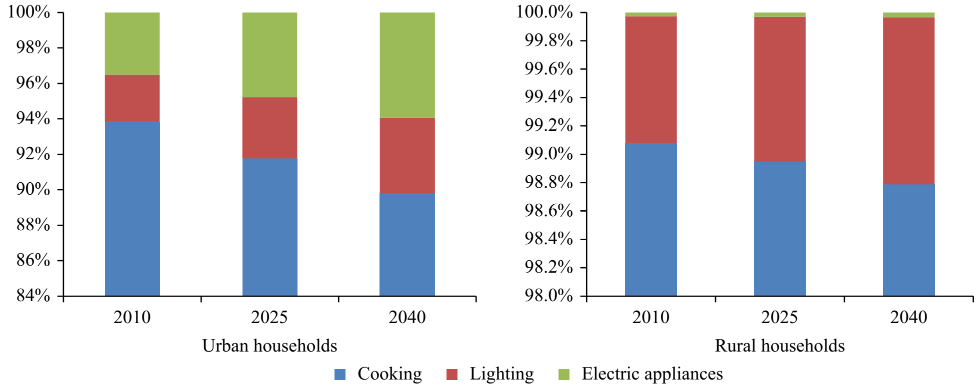

Fig. 7. Projected share of total households energy use by end-uses in urban and rural households in Nepal.

5.1% per year for lighting and 5.4% per year for electric appliances in urban households, and about 1.5% per year each for these two end-uses in rural households. While refrigerators, televisions, microwaves, iron, fan and radio are most commonly used electric appliances in urban areas, use of washing machines, computer and other electronic equipments are emerging. However, information on the breakdown of appliances energy use is not available on a consistent basis, so the analysis has focused on the aggregate level.

6.2. Region-wise per capita modern energy use

The outlook of average households' commercial energy consumption per capita and electricity consumption per capita differs markedly across the scenarios and across the regions in the country (Table 4). For example, commercial energy consumption per capita in Kathmandu valley increases from 68 kgoe in 2010 to between 133 kgoe and 186 kgoe in 2040, representing the highest increase across the regions over the study period. In contrast, people living in urban hills and *tarai* regions use less than half of the commercial energy used by people in Kathmandu valley across the scenarios over the study period. Furthermore, rural people living in all three regions (mountains, hills and *tarai*) use even less commercial energy, about one-fifth of Kathmandu valley in all scenarios over the study period. Between four scenarios, commercial energy use by both urban and rural people is the highest in SED scenario, followed by HIG, BAU and LCS scenarios in 2040. Overall, country's per capita household commercial energy use increases from 13 kgoe in 2010 to low 42 kgoe in BAU scenario and to high 50 kgoe in SED scenario in 2040.

Despite several initiatives from the government and the organizations, access to modern sources of energy in Nepal continues to be one of the lowest in the world. For example, Nepal's household commercial energy use on per capita basis is only about 5% of world average, 9% of China, 30% of India, 52% of Bangladesh and 54% of Sri Lanka in 2010 (IEA, 2012a). Furthermore, both commercial energy and electricity consumption of households in per capita in 2010 are much lower than their minimum threshold levels[4] . These alarming statistics is projected to improve only by little over the study period. For example, the national level electricity consumption per capita in 2040 is projected to be slightly above the minimum threshold level in three (BAU, HIG and SED) scenarios but it will remain below the minimum threshold for rural households over the study period. This will have a negative effect on country's economic development as majority of the population, around 69%, is projected to live in rural areas in 2040.

[4] The minimum threshold level of electricity consumption of rural households per person is assumed to be 50 kWh per year and for urban households it is 100 kWh per year (IEA, 2011). Likewise, the minimum threshold level of commercial energy consumption per person is assumed to be 100 kgoe.

Table 4
Projected household sector commercial energy and electricity consumption per capita under selected scenarios.

	Commercial* energy consumption per capita (kgoe)					Electricity consumption per capita (kWh)				
	2010	2040				2010	2040			
		BAU	HIG	SED	LCS		BAU	HIG	SED	LCS
Urban	39	89	106	110	84	165	479	560	579	340
Mountains	12	34	53	41	35	94	243	404	273	208
Hills	30	73	98	97	85	151	335	510	404	290
Kathmandu valley	68	158	159	186	133	210	796	630	904	435
Tarai	24	53	77	69	53	145	349	545	463	307
Rural	8	22	21	24	31	20	36	40	45	34
Mountains	11	23	24	20	27	12	21	23	30	21
Hills**	8	24	23	22	32	20	38	42	55	35
*Tarai***	7	20	18	26	31	21	37	40	39	35
Nepal	13	42	47	50	47	24	101	117	123	75

* Commercial energy includes both modern and alternative fuels but excludes biomass.
** For simplicity, both rural hills and rural *tarai* regions are aggregated from their five development regions.

At the regional level, Kathmandu valley dominates the outlook of electricity consumption per capita in all scenarios over the study period. The valley's per capita electricity consumption increases from 210 kWh in 2010 to high 904 kWh in SED scenario in 2040, an increase of more than four-folds. In contrast, the electricity use by majority of the rural population remains low accounting for only about 12% of urban population in 2010 and this disparity further widens to around 7% to 10% in all scenarios in 2040. Since majority of people live in rural areas, this trend of low electricity use by rural population also reflects at the national level. Indeed, the national level electricity consumption per capita is one of the lowest in the world. For example, country's projected per capita electricity use of 101 kWh in BAU scenario in 2040 is on average about the current electricity use per capita of Asia (excluding China) and Pakistan, the half of current use in China and one-fourth the current use of world average. Although, the government has a plan for 100% electrification by 2027, which is reflected in BAU scenario, the per capita electricity use in Nepal is expected to remain low unless more vigorous actions is taken.

6.3. Household cooking energy use

Unlike in developed countries, cooking is the most important energy service of Nepal and this activity consumes most of household energy use in the country. Over 98% of country's household

Table 5
Useful energy consumption *per capita for cooking under business-as-usual and three alternative scenarios (kgoe).

	2010	BAU		HIG		SED		LCS	
		2025	2040	2025	2040	2025	2040	2025	2040
Urban	48	56	56	64	61	60	62	59	63
Mountains	50	58	54	63	51	59	49	57	53
Hills	63	69	67	78	71	75	73	74	76
Kathmandu valley	32	47	56	56	65	54	65	49	59
Tarai	50	55	50	62	53	58	55	59	60
Rural	67	68	73	59	59	61	63	67	73
Mountains	73	74	79	65	66	65	64	72	75
Hills	82	84	89	73	72	73	73	82	87
Tarai	54	55	59	47	48	50	54	55	61
Nepal	64	66	68	60	60	61	62	66	70

Note: * The following conversion efficiencies are used: 15% for open fire, 20% for charcoal, 25% for ICS I and II, 55% for kerosene (pressure) stoves and 60% for LPG and biogas stoves (O'Sullivan and Barnes, 2007).

energy consumption in 2010 is used for cooking.[5] Biomass is the main fuel for cooking because it is the cheapest and most widely available fuel. Without any strong policy measures by the government to expand access to modern energy services in the country, biomass as their main fuel for cooking will continue to increase over the study period. In the BAU scenario, biomass use for cooking in the country rises from 8421 ktoe in 2010 to 10267 ktoe in 2040.

Despite the benefits of switching from traditional fuels to modern fuels for cooking, modern fuels are rarely used in rural households in the country. Several factors influence this outcome including fuel cost, lack of public awareness of new and ICS, and lack of energy infrastructure. Since it is difficult to measure energy services directly, a promising option is to measure cooking at the level of useful energy.[6] In exploring how households use energy for cooking, the amount of useful energy consumption for cooking across the regions is analyzed and presented in Table 5. Against the popular view, it is interesting to note that on average useful energy use for cooking per person in rural households is more than its use in urban households in 2010. This is due to the fact that cooking activities in rural households also involves heating their home. This is also evident from the fact that rural households spent more time for cooking than the urban households. However, there are wide variations in the level of per capita useful energy consumption for cooking across the regions. For example, useful energy consumption per capita in Kathmandu valley is 50% less of national average while it is about 25% more of national average in rural hills region in 2010. In the HIG scenario, per capita useful energy for cooking in urban areas is higher than in rural areas while it is just the opposite in BAU, SED and LCS scenarios in 2040. This is mainly due to the fact that HIG scenario exhibits both low urban population growth and higher efficiency improvement of traditional cooking devices.

6.4. Energy development indicator (EDI) and energy poverty

One of the main concerns of the Nepal's government is the high level of energy poverty situation in the country. There is a wide disparity in energy poverty level between urban and rural areas

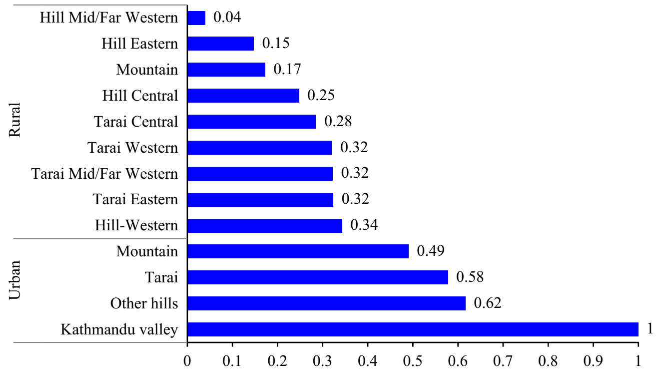

Fig. 8. Energy Development Index (EDI) by analytical regions in the business-as-usual scenario in 2040.

(Fig. 8). As expected, the urban areas have the lower energy poverty level- which is inversely related to EDI value- compared with the rural areas in 2040. Between urban areas, Kathmandu valley has the highest EDI value of 1 and urban mountains have the lowest EDI value of 0.49. This is because the valley is much more urban, and it has the highest electricity consumption per capita and access to cleaner cooking fuels of all the regions in the country. On the other hand, most of the regions in rural areas have high energy poverty levels. In particular, rural Mid/Far West development regions of hills have the lowest level of EDI because most of the households in these regions rely heavily on biomass for cooking and heating and these households have limited access to electricity. Although Mid/Far Western regions (mountains, hills and *tarai*) are the least developed regions in the country, it is interesting to note that Mid/Far Western region of *tarai* is projected to have higher value of EDI than many rural Central and Eastern development regions of the country. This suggests that government actions and plans including promotion of RETs and biomass ICS in alleviating energy poverty in the rural areas should not only be limited to Mid/Far West development regions but also should focus on Eastern, Central and Western development regions of the country.

6.5. Emission of CO_2 and local air pollutants

While most studies focus on exposure to air pollutants and its health impacts in rural areas of Nepal, a few studies illustrate region-wise inventory of air pollutant emissions. Because ecological and climatic conditions play a key role in household energy use patterns in the country, it is important to quantify these air pollution emissions at different regions. In the case of global pollutants, energy-related CO_2 emissions from the household sector in the BAU scenario is projected to rise from 373 kt in 2010 to 1392 kt by 2040, an average growth of 4.5% per year (Fig. 9). Since most of the CO_2 emissions in the country is associated with burning of fossil-fuels, urban areas including Kathmandu valley contribute more than two-thirds of total CO_2 emissions from this sector in 2010 and it is projected to increase to 82% in 2040 in BAU scenario. However, the share of rural areas in total household sector CO_2 emissions is projected to remain low due to high share of biomass, which is assumed as carbon-neutral,[7] for cooking and heating activities in these regions. Form

[5] Because of country's geography and climate conditions, energy use for cooking also involves space heating in rural households especially in hills and mountains regions. However, precise breakdown of cooking and heating in these regions is difficult and beyond the scope of this study.

[6] Measuring useful energy for cooking gives us a better idea of the energy services than measuring in primary energy or end-use energy. The quantity of end-use energy is strongly dependant on the efficiency of cooking devices.

[7] Burnt biomass emits CO_2 and other greenhouse gases. In this study, we assume biomass fuel is burnt efficiently and completely and it is replaced by new growth leading to the process as carbon-neutral. However, if the process is not considered carbon-neutral, it would be interesting to see the overall effect on CO_2 emissions of switching from biomass to modern fuels in the country which is beyond the scope of this paper.

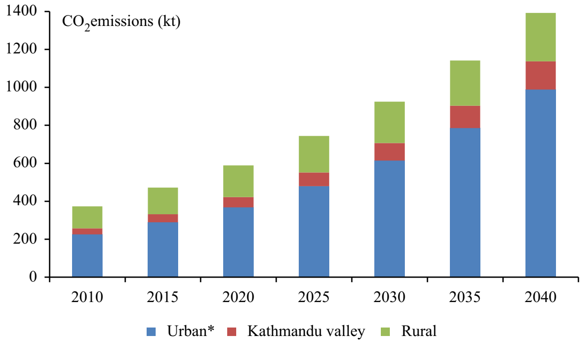

Fig. 9. Estimated CO_2 emissions from fuel combustion in the household sector under business-as-usual scenario.

Table 6
Estimated local airborne pollutants from fuel combustion in the household sector under business-as-usual scenario (kt).

	CO		PM10		SO_2		NO_x	
	2010	2040	2010	2040	2010	2040	2010	2040
Urban	508	882	29	55	3.66	5.17	0.02	0.03
Mountains	14	33	1	2	0.08	0.17	< 0.01	< 0.01
Hills	209	302	11	19	1.75	1.29	0.01	0.01
Kathmandu valley	10	21	1	2	0.44	1.96	< 0.01	< 0.01
Tarai	275	526	16	33	1.39	1.75	0.01	0.02
Rural	6971	7053	353	379	67.54	56.02	0.22	0.23
Mountains	679	681	32	34	8.37	7.19	0.02	0.02
Hills	4091	4045	194	205	48.95	40.31	0.12	0.13
Tarai	2202	2328	128	140	10.21	8.52	0.08	0.08
Nepal	7480	7936	381	433	71.20	61.19	0.23	0.26

the world's perspective, Nepal's population is only about 0.4% of the total global population and the country is responsible for only 0.025% of total greenhouse gas emissions in the world. Also, projected energy-related total CO_2 emissions and CO_2 emissions per capita of Nepal is projected to remain far below the average value of many Asian countries. For example, country's per capita CO_2 emissions from the household sector in all scenarios in 2040 will be less than the current level in Bangladesh, India and Pakistan (IEA, 2012b). Both the government and the international observers are aware of climate change concerns such as melting of glacier lakes and increase of an average maximum annual temperature by 0.06 degree Celsius in Nepal (MoEST, 2011). However, formulating and implementing any climate change-related policies in Nepal should focus more on adaptation and less on mitigation. In this regards, establishment of Climate Change Council in 2009 and its preparation of National Adaption Program Action (NAPA) a year later, and adoption of climate change policy 2011 are some of the commitments shown by Nepal's government to tackle climate change concerns in the country.

Unlike CO_2 emissions, exposure to indoor local air pollutants from burning solid fuels for cooking and heating is widespread in Nepal. Exposure to these emissions has negative health impacts and these emissions carry a high social and economic cost. Despite these facts, efforts to mitigate local indoor air pollutant emissions in the country both by the government and the organizations involved in Nepal's development have limited success. Although several factors affect exposure to indoor air pollution, including fuel and stove type, ventilation, and amount of time spent indoor, it is clear that source of the pollution has been the most important factors. This study finds that local air pollutants are significantly higher in rural areas than in urban areas and they are projected to follow the similar path over the study period (Table 6). This is because rural households are more dependent on biomass and they rely on low-quality cooking equipments. In addition, most of the rural households are poorly ventilated and their cooking activities also involve heating especially in hills and mountains regions of the country. These factors often exacerbate negative health impact on rural households than on urban households, as there is incomplete combustion and non-dissipation of smoke.

Despite country's total CO emissions is projected to increase only by 6% in next 30 years, more than 88% of these emissions in BAU scenario in 2040 come from rural households (Table 6). In particular, rural hills and *tarai* regions together, where majority of the rural population is concentrated, contribute 90% of total rural CO emissions and 80% of national CO emissions in 2040. In terms of contribution to total urban CO emissions, the share of *tarai* region is projected to be the highest (60%), followed by hills (34%), mountains (4%) and Kathmandu valley (2%). Likewise, emissions of PM10 follow the similar trend reaching 433 kt in 2040 compared to 381 kt in 2010, with more than 87% of these emissions in 2040 are coming from rural households. In contrast, SO_2 emissions are projected to decrease from 71.2 kt in 2010 to about 61.2 kt in 2040 due to declining use of kerosene and charcoal over the study period. Almost 95% of total SO_2 emissions are coming from rural areas in 2010 and this share is projected to decrease to 92% in 2040. Oxides of nitrogen (NO_x) are also formed during combustion but increase in these emissions, from 0.23 kt in 2010 to 0.26 kt in 2040, is less significant in the future. Similar to the SO_2 emissions trends, rural households contribute over 92% in 2010 and 88% in 2040 of total NO_x emissions in BAU scenario.

It should be noted that there is uncertainty inherent to the methodology and assumptions in the study. For example, if the range of socio-demographic and technological parameters changes is larger than in the four scenarios considered in the study, influences on projected household energy demand and corresponding environmental implications could be much different. Furthermore, all scenarios considered in the study are non-intervention scenarios and any explicit changes or intervention by the government in the future could also likely change the results.

7. Conclusion

Unlike in developed countries, the household sector accounts for most of the energy consumed in Nepal and this trend is projected to remain the same in next 30 years. Despite this high share, both electricity- and commercial energy- consumption by Nepalese people remain below the basic human needs and they are one of the lowest in the world. Furthermore, the EDI – a measure of energy poverty level – of Nepal is at the bottom of rankings compared to selected developing countries of Asia, reflecting country's low rates of household electrification and high share of biomass use for cooking. In addition, the local air pollutant emissions from burning biomass for cooking in Nepal and its negative health impact, especially women and children in rural areas, are quite significant. Although the share of rural households decreases from 80% in 2010 to 65% in 2040, these households play an important role in country's energy system development. Therefore, increasing access to clean, affordable and reliable energy services particularly to rural households is essential in reducing energy poverty and improving indoor air quality in Nepal. If the present trend continues without any new policy initiatives, country's energy poverty situation in the future would improve only by little. This limited improvement in energy poverty situation will have significant implications for country's social and economic development.

There are significant regional and geographical variations as well as differences between urban and rural areas in the patterns of household energy consumption in Nepal. These regional differences in energy use have important implications for policy initiatives. Although the share of biomass in total household energy demand is projected to decline in future, these fuels will remain the primary energy sources of households in the country. Specifically, rural areas will remain highly dependent on biomass energy, mainly fuelwood, for cooking. Compared to national average, urban residents use three times as much as commercial energy per capita and close to seven times as much as electricity consumption per capita. Within urban areas, the residents of Kathmandu valley dominate both commercial energy use and electricity consumption due to higher average incomes and better access to modern energy services. By contrast, useful energy consumption per capita for cooking of urban residents is slightly lower than the national average. At the region level, both rural hills and rural *tarai* of Mid/Far Western development regions have one of the worst energy poverty situations in the country.

Transition from biomass energy to cleaner fuels, such as LPG and biogas, for cooking, development and deployment of indigenous and innovative renewable energy technologies, and improvement of supply- and demand-side energy efficiency are some of the most cost-effective measures in alleviating energy poverty and improving access to energy in the country. Though they are straightforward, Nepal's government in coordination with inter-agencies and in collaboration with overseas development agencies should initiate plan and policies to implement these measures. These initiatives should include identifying the needs for the poor, formulating and implementing policies in developing community based local energy resources, developing private–public partnerships in promoting RETs, promoting advanced and improved

Table A1
Estimated household sector final energy consumption by analytical regions in 2010 (ktoe).

Analytical region	Cooking							Lighting				Others		Total
	Traditional			Commercial**			Rene wables	Commercial			Rene wables	Commercial		
	Fuel wood*	Agriculture residue	Animal dung	KERO	LPG	ELE	Biogas	KERO	ELE	OTH	Solar	ELE	OTH	
Urban	672	30	45	3.5	92.3	12.3	13.8	0.9	21.7	–	0.08	30.2	–	922
Mountains	18	1	1	0.0	0.1	0.1	0.2	0.0	0.4	–	0.01	0.2	–	21
Kathmandu valley	69	–	–	1.7	66.6	6.3	0.6	–	7.7	–	.01	10.9	–	163
Hills others	233	9	16	0.6	11.3	2.2	5.6	0.1	4.9	–	0.04	6.6	–	289
Tarai	352	20	28	1.1	14.2	3.8	7.3	0.7	8.7	–	0.02	12.4	–	449
Rural	6904	294	514	–	6.3	–	51.4	30.5	36.9	3.1	0.17	1.0	1.0	7843
Mountains	564	16	58	–	0.5	–	10.1	2.9	1.7	0.3	0.00	0.0	0.1	653
Hills eastern	839	22	49	–	0.6	–	2.5	4.9	2.8	0.5	0.03	0.1	0.2	921
Hills central	1189	26	61	–	0.8	–	6.7	4.5	6.3	0.5	0.02	0.2	0.2	1295
Hills western	728	38	77	–	0.6	–	7.4	2.1	4.8	0.2	0.03	0.1	0.1	858
Hills M/F western	810	19	74	–	0.6	–	2.5	6.7	1.3	0.7	0.04	0.1	0.2	915
Tarai eastern	652	48	63	–	0.8	–	4.8	2.2	3.0	0.2	0.01	0.1	0.1	774
Tarai central	925	59	46	–	1.0	–	3.4	3.4	6.5	0.3	0.01	0.2	0.1	1045
Tarai western	566	31	32	–	0.7	–	7.4	1.8	5.1	0.2	0.02	0.1	0.1	644
Tarai M/F western	630	37	54	–	0.7	–	6.5	2.1	5.4	0.2	0.02	0.1	0.1	736
Total	7576	324	560	3.5	98.5	12.3	65.2	31.4	58.6	3.1	0.25	31.2	1.0	8765

* Fuelwood includes charcoal.
** KERO is kerosene, LPG is liquefied petroleum gas, ELE is electricity and OTH is others (i.e., candles and dry cells).
Sources: ADB (2004), AEPC (2012a), IEA (2012a), MoF (2012), NEA (2011), NOC (2013), World Bank (2012), WECS (2010).

Table A2
Summary of driving factors and main features of four scenarios.

Scenario	Demography* (Average annual growth rate)			Technology (In every 10 years)			Access to electricity (In every 10 years)		Fuel substitution (cooking) (In every 10 years)		
	Urban (%)	Rural (%)	Period	Urban (%)	Rural (%)	End-use	Urban** (%)	Rural (%)	Urban (%)	Rural (%)	Fuel type
BAU	4.16	1.39	2010–20	+1.5	+1.25	Lighting	100	100***	2	1	LPG
	3.83	1.08	2020–30	−0.75	−0.75	Cooking			1	2	ICS
	3.47	0.48	2030–40						0.5	0.25	Electricity
									0.5	1	Biogas
HIG	4.09	1.32	2010–20	+2	+1.5	Lighting	100	5	2.25	1.15	LPG
	3.42	0.68	2020–30	−1.5	−1.5	Cooking			1	2	ICS
	3.03	0.06	2030–40						1	0.5	Electricity
									0.5	1	Biogas
SED	4.09	1.32	2010–20	+1.5	+1	Lighting	100	5	1.5	0.75	LPG
	3.42	0.68	2020–30	−1.5	−1.5	Cooking			1.5	3	ICS
	3.03	0.06	2030–40						0.5	0.25	Electricity
									0.5	1.5	Biogas
LCS	4.16	1.39	2010–20	+1.25	1	Lighting	100	4.5	1	0.5	LPG
	3.83	1.08	2020–30	−1.125	−1.125	Cooking			2	4	ICS
	3.47	0.48	2030–40						0.75	0.4	Electricity
									1	2	Biogas

* Based on UN's medium and low variant population projection.
** By 2020, all urban households have access to electricity.
*** By 2027, all rural households have access to electricity.

biomass cookstoves, and scaling up solar, biogas and ICS programs. Due to geographic diversity, the government should also focus on financing and involving private sector in developing community owned off-grid micro hydro system in the country. To achieve these goals, the government must develop a sound system of collecting and monitoring periodic energy-related data at different regional levels and make these data available for general public.

Further works that explores the potential for changes in household characteristics such as size, age, and composition in more detail would be worthwhile on understanding of evolution of household fuel choice and adoption of ICS. In addition, a better understanding of behavioral components of households can improve projections of region-wise energy demand in more detail.

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Appendix A

See Tables A1 and A2

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